

ANA / HEp-2 Immunofluorescence Pattern Classification

Complete working context as of 18 September 2026. Paste or attach at the start of a new session to bring an assistant fully up to speed.

1 Project Identity and People

Project title (final, do not change): Automated Classification of ANA Immunofluorescence Patterns Using Deep Learning.

One-line description: build and evaluate a deep learning system that classifies ANA immunofluorescence staining patterns from HEp-2 cell images, using both public benchmark datasets and a private clinical dataset from Indus Laboratory, Karachi.

Who	Role
Muhammad Yousuf	FYP group leader, roll no. 23K-0690, FAST-NUCES Karachi
Arsalan Ahmed	FYP group member, roll no. 23K-0536
Muhammad Huzaifa Altaf	FYP group member, roll no. 23K-0627
Dr. Muhammad Shahzad	Supervisor. Would be Principal Investigator on any HEC grant application.
Dr. Sabiha Anis	Consultant, Clinical Immunology, Indus Laboratory, Karachi. Clinical collaborator, source of data and expert annotation.
Dr. Fahim	Clinical side, a colleague of Dr. Sabiha in his own right, not her assistant. Handles IRB template and process.
GU Tech	The project also runs as an internship under GU Tech. In documents produced under that banner the students are referred to collectively as “the intern team, GU Tech” and individual names are not written.

Naming convention. Two parallel document streams exist. The **GU Tech / synopsis / NRPU** stream credits work to “the intern team, GU Tech, under Dr. Muhammad Shahzad” with no individual names. The **FAST FYP** stream names all three students with roll numbers. Do not mix the two conventions.

2 Clinical Domain Primer

- Autoimmune disease** → the immune system produces **autoantibodies** that attack the body’s own cells.
- The **ANA test** (antinuclear antibody) is the standard screen, performed by **IIF** (indirect immunofluorescence) on **HEp-2 cells**, a standardised human cell line used as a substrate. HEp-2 = Human Epithelial type 2. Nothing to do with hepatitis.
- Patient serum is applied to the slide; bound autoantibodies are tagged with a fluorescent dye and glow under a microscope.
- Different autoantibodies bind different cell structures, producing visually distinct **staining patterns**. The pattern indicates likely target antigens and guides which follow-up antibody test to order.
- ICAP** = International Consensus on ANA Patterns, the naming standard. Each pattern has an **AC code**. The scheme runs **AC-0 to AC-31** — 32 coded categories, of which 31 are positive staining patterns (AC-0 = negative). Split into **competent-level** (11 reportable groups a routine lab must recognise) and **expert-level**.
- The problem:** interpretation is manual, slow and subjective. Experienced readers agree well with each other ($\kappa = 0.849$) but a novice agrees only moderately ($\kappa = 0.528$).
- The output is a pattern, not a diagnosis.** Never claim the system diagnoses disease.

Pattern frequency at Indus Laboratory

Common: **AC-1 (homogeneous)**, the **speckled group** (AC-2, AC-4, AC-5) and the **nucleolar group** (AC-8, AC-9, AC-10). Rare: **AC-3 (centromere)**. Class imbalance will be real. The project therefore starts with the frequent patterns and extends to the rest later.

3 Literature Reviewed — Four Papers

#	Paper	Purpose	Key result
1	Rahman et al. 2020, Med Image Anal 65:101764	Review of 31 DL papers to Oct 2019	Cell-level on I3A saturated ~99%
2	Wu et al. 2021, Diagnostics 11(4):642	Model for full ICAP standard, 11 groups + mixed	$\kappa = 0.844$ vs experts
3	Mayr et al. 2025, J Clin Med 14(19):6924	Commercial system on routine samples	$\kappa = 0.69$ pos/neg
4	Jain et al. 2026, EULAR Rheum Open 2(1):111-118	Hierarchical staged classification	87.16% → 92.27%

Rahman (review). Two axes: cell-level vs specimen-level input; DNN as feature extractor vs end-to-end classifier. Four public datasets (ICPR 2012/MIVIA, SNPHEP-2, I3A Task-1/2, AIDA). Metrics ACA vs MCA, LOSO protocol. Findings: rotation augmentation essential (cells have no canonical orientation), GAN augmentation lost to plain rotation, keeping cell background helps, deeper is not better (HEp-Net matches DRI-Net at 7% size). Open challenges: imaging variation across labs (their pick for the real problem), class imbalance, labelling cost, explainability, result consistency.

Wu. Reframed as **multi-label** (sigmoid + BCE, not softmax). 34,756 samples, one hospital in Taiwan, temporal split (train 2018-19, test 2020). Compared 6 CNNs; InceptionResNetV2 won (F1 = 0.86, κ = 0.82). Image→sample voting with two hard-coded clinical exception rules. Headline: κ = 0.849 expert-expert, **0.844 expert-AI**, 0.528 expert-novice.

Mayr. 143 **consecutive** routine samples, not curated. Tests pos/neg, titre, pattern. Findings: missed 21 of 94 human-positives (too conservative, low-titre); titre over-estimated, clustering at 1:10,240; pattern κ = 0.54 → 0.67 same substrate → 0.80 ignoring AC-2 vs AC-4/5. Methodological catch: human arm used AESKU slides, AI arm used Medipan; they ran a same-substrate control.

Jain. hmsANA hierarchical pipeline: Level 0 pos/neg (98.02%) → Level 1 metaphase stained? (96.57%) → Level 2A AC-1/2/3 (87.22%) or 2B AC-4/5 (92.35%). Overall 92.27% vs 87.16% flat, a gain of roughly 5.1 percentage points. 803 samples filtered from 3,712 (mixed patterns removed). YOLOv8 hugely outperformed rivals on hard pairs (92.35% vs 55.29% for InceptionResNetV2 on AC-4 vs AC-5).

The synthesis / research gap. Wu has the scope, Mayr has the realism, Jain has the structure. No system has all three. All four are single-centre with no external validation. Note that the flat-vs-staged comparison *has* been done by Jain — the gap is that it was one curated dataset, five patterns, using a metaphase-plate split that does not extend to the wider ICAP set. Do not write as though the comparison has never been attempted.

4 Meeting at Indus Laboratory (1 September 2026)

Attendees: Dr. Sabiha Anis, Dr. Fahim, Dr. Muhammad Shahzad and the intern team. Approximately one hour. The four-paper literature review deck was presented in full; Dr. Sabiha then walked through the ICAP patterns using anapatterns.org.

Key points from Dr. Sabiha

- ICAP defines 32 patterns including mixed patterns, far more than the 5–11 used in most papers.
- AC-1, speckled and nucleolar are common at Indus; AC-3 (centromere) is rare. Class imbalance will be real.
- Similar-looking patterns are the main clinical challenge.
- Data volume: approximately **480 images per month**; each patient sample yields **3-4 microscopy images**.
- She can help source data from other clinics and hospitals.
- **Data is confidential.** Work may need to be local. Controlled Google Drive access was discussed but left **unresolved**; Dr. Shahzad to follow up.
- **IRB approval is mandatory** before using any clinical data. Dr. Fahim to provide the template.
- Dr. Sabiha and Dr. Fahim to provide a list of the most common ANA patterns at Indus.
- PAP Conference 11–13 December mentioned.

What Dr. Shahzad proposed

- Start with binary ANA positive vs negative, following Jain’s staged approach.
- Do not attempt all 32 patterns; focus on the most frequent ones first.
- Begin with public datasets and pretrained models while IRB is pending.
- Long term: a web application where a clinician uploads an image and gets a predicted pattern.

Agreed phases: IRB proposal → data acquisition → pos/neg classification → pattern classification → validation → deployment.

5 Public Dataset Landscape

Already obtained

Dataset	Level	Classes	Size and notes
AIDA_HEp-2	Specimen	22 ANA patterns, ICAP nomenclature, single and multiple patterns per image	2,080 images — 1,498 positive, 582 negative. The only widely available public set containing negatives , which makes it the key resource for the positive-vs-negative stage. Reference: Benammar Elgaaiet et al., BioMed Research International, 2016.
I3A Task-1	Cell	6: Homogeneous, Speckled, Nucleolar, Centromere, Golgi, Nuclear Membrane	13,596 cell images from 83 specimens. The most widely used comparison benchmark in published work.

To be obtained

Dataset	Access	Notes
ICPR 2012 / MIVIA	Request form at mivia.unisa.it/datasets-request	1,455 cell images from 28 specimens, 721 train / 734 test. 6 classes incl. Cytoplasmic. Sample zip available without registration at mivia.unisa.it/database/biomedical/Example.zip
SNPHEp-2	Direct download, no gate	1,884 cell images from 40 specimens, 5 classes, 3 MB. http://dropbox.eait.uq.edu.au/uqawilie/SNPHEp2/SNPHEp2_images.zip — NICTA licence, non-commercial research only, must cite Wiliem et al. WACV 2013
I3A Tasks 2, 3, 4	Same gate as Task-1, hep2.unisa.it/access.html	Task-2 specimen-level (237 images, ~48,627 cells); Task-3 mitotic cell recognition; Task-4 cell segmentation
BTU Cottbus cell-cycle set	Open, CC-BY. zenodo.org/records/18337703	100,000 DAPI-stained single-cell images at 256×256, labelled interphase / prophase / metaphase / anaphase / telophase. Not ANA patterns, but directly relevant to the metaphase-split question
Zhejiang fine-grained set	Contact authors	Ma et al., Med Image Anal 2026. ~40,000 images, 17 ICAP basic patterns + 5 combinations. Closest thing to full-ICAP coverage that exists
Oulu HEp-2 set	Open Dropbox, qixianbiao.github.io/HEp2Cell	Undocumented size and classes. Supplementary to a 2015 hand-crafted-descriptor paper. Frictionless but unverified provenance — do not cite as a primary source
HuggingFace / Kaggle mirrors	MuGeminorum/HEp2 , Genius-Society/HEp2 , dumplinghead/hep-2-cells	Unofficial re-hosts, apparently of I3A Task-1. Convenient for fast iteration; cite the originals in any publication

Known to exist but not obtainable

Wu's Taichung set, Jain's hmsANA set, Mayr's routine samples, the Changsha dataset, the Tunis military hospital set, the Zhejiang 2023 Bioinformatics sets and the Xinhua Hospital corpus. All single-centre, all private.

Two structural facts. (1) No public set covers the full ICAP AC-0-31 taxonomy; legacy class names like “Fine Speckled” and “Coarse Speckled” are not ICAP categories, so label mapping is required before merging. (2) Outside AIDA, public sets are positive-only, so the positive-vs-negative stage depends on AIDA plus the local clinical data.

6 Documents Produced So Far

File	What it is
HEp2_Review_Summary.pptx	12 slides — Paper 1 (Rahman)
ICAP_ANA_ML_Summary.pptx	12 slides — Wu, labelled “Paper 3 of 4”
AI_ANA_Pilot_Study_Summary.pptx	12 slides — Mayr, labelled “Paper 2 of 4”
ANA_HEp2_Literature_Review.pptx	26 slides — consolidated review of all four, for the supervisor
Meeting_Minutes.pdf	1 page — MoM from the Indus meeting, dated 1 September 2026
Project_Synopsis.pdf	4 pages — the formal synopsis, dated 9 September 2026
FYP-I_Project_Proposal.docx	The FAST FYP-I proposal, current working deliverable
NRPU_Stage1_Outline.pdf	5 pages — draft Stage-I outline for the HEC NRPU 2026-27 grant call

7 Style and Formatting Preferences

General. Python. Direct, no-fluff communication. For build guidance: complete copy-paste-ready code and sequential steps. Documents must be concise — short bullets over prose paragraphs, because people will not read long documents. Language in documents should read naturally and not sound machine-generated.

The four PPTX decks — dark theme

Deep green-black background [#0D1F1B](#), card [#17352E](#), alt-card [#1F453B](#), fluorescence-green accent [#3DDC84](#), pale mint [#7FD8BE](#), text [#EAF6F1](#), muted [#94B3A9](#). Cambria headings, Calibri body. Every slide has speaker notes.

The synopsis and other PDFs — light navy + copper theme

Green was explicitly rejected for documents. Primary navy [#17395C](#), copper accent [#C2703C](#), tint [#F4F7FA](#), alt tint [#E5EFEB](#), muted [#5A6B7B](#), rules [#D3DEE8](#), ink [#1F2933](#). DejaVu Sans body, justified text, section headings with copper numerals. Navy table headers with

white text, zebra-striped rows. Light callout boxes with a copper left bar; dark navy callouts for key statements. Generated with **wkhtmltopdf** from **hand-written HTML**, not reportlab.

The FAST FYP proposal — university template

Times New Roman throughout, A4, 1" margins on all sides. H1 16pt bold, H2 13pt bold, H3 12pt bold italic, body 12pt, single line spacing, justified. Figure captions below figures at 11pt centred; table captions above tables. IEEE referencing.

8 Locked Decisions — Do Not Regress These

- **Everything is tentative except data acquisition.** In the synopsis, methodology, tools and phases each carry a "(Tentative)" tag — but only **three** such tags total. Hedging was deliberately cut back; do not add "to be decided" everywhere.
- **Flat vs staged is NOT decided.** Both will be implemented and compared. Never write as though the staged design is chosen.
- **Multi-label / mixed patterns are OUT of scope.** This was added once and then removed. Documents are single-label throughout. May be revisited much later.
- **FastAPI is confirmed** as backend, but list it plainly in the tools table, do not make a feature of it. **Do not use the term "REST API"** anywhere; say "web application".
- Diagrams must be **general** (k groups), not locked to two branches.
- Splits must be made at **sample level, not image level**, since 3–4 images come from one patient.
- Synopsis date: 9 September 2026. MoM date: 1 September 2026.

Figures in the synopsis

Two embedded PNGs, cropped and flattened to white before embedding. **Figure 1** — flat classifier: HEp-2 Image → Preprocessing → CNN Backbone → Softmax (N classes) → One Label. **Figure 2** — staged pipeline in general form: Images → Preprocessing → STAGE 1 (pos/neg, negative branch exits to "Report Negative") → STAGE 2 (broad group assignment) → STAGE 3.1 / 3.2 / 3.k (one classifier per group, 3.k dashed to show extensibility) → Image-to-Sample Aggregation.

9 Citation Corrections Already Applied

1. **Mayr author list** — correct form is Mayr, S., Dollinger, M., **Ehrenstein, B.**, Günther, F., Krammer, O., Schuster, A., Büttner, T., Hiemann, R., Schierack, P., Roggenbuck, D., & Fleck, M. An earlier version had a corrupted "Merenstein, B."
2. **AC-30 must not be cited to Chan et al. 2022** (6th workshop). AC-30 was introduced at the 7th workshop. Cite Andrade et al. 2024 plus the ICAP site.
3. **Pattern count** is phrased as "AC-0 to AC-31 — 32 coded categories, of which 31 are positive staining patterns", not bare "32 patterns", because AC-0 is negative and not a pattern.
4. Claims are deliberately softened: "performance on routine samples **remains more challenging than** results on curated benchmarks" (not "drops"); "**many** published systems are single-centre" (not "nearly all").
5. ICAP web reference carries an access date.

Verified-correct numbers, safe to cite: $\kappa = 0.849 / 0.844 / 0.528$ (Wu); up to four mixed patterns (Wu); 87.16% → 92.27% (Jain); AC-2 vs AC-4/5 dominates Mayr's pattern error; AIDA 2,080 images with 1,498 positive and 582 negative; I3A Task-1 13,596 cell images from 83 specimens.

10 FYP-I Proposal — Current State

Built to the FAST FYP-I template. Registration table marks the project as Industrial / Research & Development, SDGs Good Health and Well-Being plus Industry Innovation and Infrastructure, specialisation AI + Data Science + Web Development.

Section structure (numbering matches the template)

1 Project Abstract · 2 Introduction (2.1 Research Domain, 2.2 Importance of the Domain, 2.3 Current Situation, 2.4 Relevance of Computational Research) · 3 Problem Statement · 4 Motivation (4.1 Aims and Objectives, 4.2 Scope of the Project with 4.2.1 In Scope and 4.2.2 Out of Scope) · 5 Proposed Methodology (5.1 Overview of the Two Candidate Designs, 5.2 Data Acquisition, 5.3 Image Preprocessing, 5.4 Candidate Staged Pipeline, 5.5 Model Development and Transfer Learning, 5.6 Image-to-Sample Aggregation, 5.7 Evaluation Protocol, 5.8 Ethical Considerations) · 6 Functional and Non-Functional Requirements (6.1, 6.2) · 7 System Architecture (7.1 Presentation Layer, 7.2 Application Layer, 7.3 Data Layer) · 8 Individual Tasks · 9 Gantt Chart · 10 Tools and Technologies · 11 Expected Deliverables (11.1 FYP-I Semester 7, 11.2 FYP-II Semester 8) · 12 References.

Objectives O1–O7

O1 build the dataset · O2 image-level pos/neg model · O3 pattern classifier starting with frequent patterns · O4 compare flat vs staged · O5 image-to-sample aggregation · O6 evaluate on imbalance-appropriate metrics and measure expert agreement · O7 deploy behind a web application.

Deliverables split across the two semesters

FYP-I (Semester 7): literature review and domain study; public datasets prepared (AIDA + I3A Task-1 and any further sets); IRB proposal submitted and data-sharing arrangement agreed; preprocessing and augmentation pipeline implemented; positive-vs-negative model trained and evaluated on public data; initial architecture comparison for that stage; SRS, initial design documentation and the FYP-I report.

FYP-II (Semester 8): local clinical dataset integrated and label-aligned; pattern classifier across the prioritised class set; both architectures implemented and compared with per-pattern breakdown; image-to-sample aggregation scheme; agreement with expert reporting measured; web application; test reports, documentation and final report.

Placeholders still to be filled

- CGPA for all three members
- Phone numbers for all three members
- Signatures (handled on the printed copy)
- Project ID — leave blank, marked “for office use”

To verify before submission

- Emails `k230536@nu.edu.pk` and `k230627@nu.edu.pk` were derived from the leader’s pattern and need confirming
- Whether the collaborator is correctly named “Indus Laboratory, Karachi” or “Indus Hospital” — earlier documents use Indus Laboratory
- Registration checkbox selections, with the supervisor
- The date currently reads 1 September 2026, carried from the template
- Whether a supervisor row is expected in the registration table

Team task allocation as written

Muhammad Yousuf — literature review and clinical domain study; IRB proposal; staged pipeline design and implementation; comparative evaluation; final report. **Arsalan Ahmed** — public dataset acquisition and organisation; preprocessing and augmentation pipeline; pos/neg baseline model; flat classifier implementation; architecture comparison and tuning. **Muhammad Huzaifa Altaf** — clinical dataset integration and annotation alignment; image-to-sample aggregation; validation against expert reporting and error analysis; web application and model deployment; UI testing and integration.

11 HEC NRP 2026-27 Grant Application

NRP is the Higher Education Commission’s main competitive research funding programme for Pakistani universities. Dr. Shahzad intends to apply using this project. He handles all formalities — eligibility, ORIC endorsement, the Rs. 5,000 fee, institutional matters. The students’ contribution is the written content.

Mechanics

- **Stage-I project outline due Friday 30 October 2026.** Short document, blind-reviewed by a panel of subject specialists. Only shortlisted outlines are invited to Stage II.
- **Stage II** is the full detailed proposal with budget, CVs, equipment quotations.
- Portal: `research.hec.gov.pk`. Endorsement by the sponsoring institution’s ORIC is required; applications from collaborating institutions are not entertained.
- PI must be a faculty member with a PhD or FCPS/MCPS/FRCS, on regular or contract basis for at least the project duration. A PI can hold only one NRP grant at a time.
- **Maximum duration 24 months.** Basic Research Grant up to PKR 20 million; Thematic Applied can exceed that with industry contribution.
- Disbursement for a 24-month project: 60% / 30% / 10%, with the final 10% covering only honoraria and overheads.

Domain choice and reasoning

The 2024 policy splits funding into Basic Research (30% of funds, any discipline, 20M cap) and Commission’s Strategic Research (70%), the latter subdivided into Thematic Applied Research across seven themes and Solicited Applied Research (invitation only, so unavailable). **Thematic Applied Research — Health, Wellbeing and Biotechnology** was chosen because that theme explicitly lists diagnostic kits and diseases specific to the geography of Pakistan, it draws on the larger funding pool, and it rewards academia-industry collaboration, which the Indus partnership provides. Digital Technologies and Cyber-Physical Systems (which lists AI) was the alternative; Health was preferred because the outcome is a diagnostic capability and AI is the method, so clinical reviewers will grasp the significance faster.

Open question. The 2026-27 poster advertises three domains — Social/Behavioral/Economic & Human Sciences, Natural/Emerging Sciences, Product/Process/Technology Development — which do not match the 2024 policy’s seven themes. The actual 2026-27 call document is needed to confirm which scheme applies. If the poster’s three are the live options, the project would sit under Natural/Emerging Sciences or Product/Process/Technology Development.

Stage-I outline structure (from the 2024 policy, section 3)

1 Problem statement · 2 Objectives and alignment with national priorities · 3 Work already done by the PI or others · 4 Project team including internal and external collaborators · 5 Plan for gaining commitment from proposed partners · 6 Beneficiaries and what the study delivers that would not otherwise be realised · 7 Identification of outcomes · 8 Sustainability plan beyond the grant duration.

Sections 5, 6 and 8 do not exist in the synopsis and were written fresh. A hypothesis was also added, since a missing hypothesis is a known rejection reason.

Financial rules worth knowing

- Students engageable as Research Assistants: PhD Rs. 80,000/month, MS/MPhil Rs. 40,000/month, undergraduate Rs. 20,000/month. Only the university's own students.
- Project staff: Research Assistant (BS/MS) Rs. 100,000/month, Research Associate (fresh PhD) Rs. 130,000/month.
- PI honorarium per year by designation: Assistant Professor Rs. 150,000, Associate Rs. 175,000, Professor Rs. 200,000. Co-PI gets one on completion at the same rate.
- Equipment and consumables together must not exceed 70% of direct cost. The university must certify in writing that it does not already have access to the equipment.
- Local travel maximum Rs. 150,000/year. International travel allowed with prior HEC approval — one fieldwork visit per year up to 30 days, one conference visit per year up to 6 days.
- University overheads in three tiers: 5% basic, 10% extra, 15% performance.
- **Private-sector collaborators receive no financial transactions.** Indus contributes in kind — data, expert annotation, validation time.

Compliance risk, effective 1 August 2026. Objectives, scope and methodology approved at award cannot subsequently be changed. Late reporting forfeits honoraria and ORIC overheads for the delayed period. Failure to submit both the annual and final reports past the three-month extension makes the project “abnormally delayed”: full refund of everything disbursed plus bank profit, a letter of displeasure, and the PI barred from submitting any HEC proposal for five years, with names published on the HEC website. Institutions with five or more such projects are ineligible for new grants.

Practical consequence: the grant proposal must commit to the flat-vs-staged *comparison* as the methodology, never to “we will decide the architecture later”. The comparison is a fixed method with a defined deliverable; an undecided architecture is a scope change waiting to happen.

12 Hardware Requirements

NVIDIA GPU with 12–16 GB VRAM (RTX 3090/4090 or A100), or cloud GPU (Colab Pro / Kaggle) for public data only. 500 GB to 1 TB storage. 32 GB RAM. A secure access-controlled workstation for the clinical dataset. Clinical data must not go to shared or public cloud storage.

13 Open Items and Next Steps

Short term

1. Continue collecting public datasets; begin exploration and preprocessing on AIDA and I3A Task-1. SNPHEP-2 needs no approval and can be pulled immediately; MIVIA and the remaining I3A tasks need request forms submitted.
2. Prepare the draft IRB proposal with a clearly stated methodology, using the template from Dr. Fahim.
3. Receive the list of most common ANA patterns at Indus to fix the initial class set.
4. Obtain the actual 2026–27 NRPU call document and Stage-I form to confirm domain categories and length limits.
5. Secure a written letter of collaboration from Indus Laboratory confirming the in-kind contribution.

Medium term

6. Baseline experiments with pretrained models on public data; implement ANA positive vs negative.
7. Confirm the data-sharing arrangement, on-site versus controlled Drive access — Dr. Shahzad.

Ongoing

8. Dr. Sabiha exploring additional data from other clinics and hospitals.

Still undecided

- Flat vs staged architecture
- Deep learning framework (PyTorch vs TensorFlow/Keras)
- Front-end framework
- Whether and when to attempt mixed patterns
- Exact Stage 2 group boundaries — candidates are the ICAP top level (nuclear / cytoplasmic / mitotic) or metaphase-plate staining as Jain used

14 Gotchas Learned Along the Way

- **AC-2 and AC-30 both stain the metaphase plate.** If Stage 2 splits on metaphase staining as Jain did, they land in the same branch and still need separating downstream.
- Jain's metaphase split works cleanly for AC-1 to AC-5 only. It does **not** generalise to the full ICAP set. The ICAP tree's own top level (nuclear / cytoplasmic / mitotic) is the more natural split for a wider class set.
- I3A's official test sets were never released, so accuracy figures across published papers are **not strictly comparable**.
- Jain's rods-and-rings F1 = 1.00 came from a heavily augmented tiny class. Do not present it as a genuine strength.

- Splits should be made at **sample level, not image level**, since 3–4 images come from one patient.
- Legacy public-dataset class names (“Fine Speckled”, “Coarse Speckled”) are not ICAP categories. Label mapping is needed before merging datasets.
- The AIDA project website currently returns 404 on the pages that published papers cite. The dataset has been obtained, but the public link should not be given as a live reference without checking.